Definition: If $v_1, \dots, v_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $v_1, \dots, v_p$ is denoted by Span$\{v_1, \dots, v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated) by $v_1, \dots, v_p$**. That is, Span$\{v_1, \dots, v_p\}$ is the collection of all vectors that can be written in the form $\boxed{c_1v_1 + c_2v_2 + \dots + c_pv_p}$ with $c_1, \dots, c_p$ scalars. $\square$

Asking whether a vector b is in Span$\{v_1, v_2, \dots, v_p\}$ amounts to asking whether the vector equation $\boxed{x_1v_1 + x_2v_2 + \dots + x_pv_p = b}$ has a solution, or, equivalently, asking whether the linear system with augmented matrix $[v_1\ v_2 \dots v_p\ b]$ has a solution.

Note that Span$\{v_1, \dots, v_p\}$ contains every scalar multiple of v_1 (for example), since $cv_1 = cv_1 + 0v_2 + \dots + 0v_p$. In particular, the zero vector must be in Span$\{v_1, \dots, v_p\}$.

A Geometric Description of Span$\{v\}$ and Span$\{u, v\}$

Let v be a nonzero vector in $\mathbb{R}^3$. Then Span$\{v\}$ is the set of all scalar multiples of v, which is the set of points on the line in $\mathbb{R}^3$ through v and 0. See Figure 10.

If u and v are nonzero vectors in $\mathbb{R}^3$, with v not a multiple of u, then Span$\{u, v\}$ is the plane in $\mathbb{R}^3$ that contains u, v and 0. In particular, Span$\{u, v\}$ contains the line in $\mathbb{R}^3$ through u and 0 and the line through v and 0. See Figure 11.

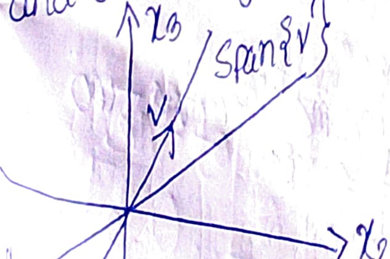

Fig 10: Span$\{v\}$ as a line through the origin

Fig 11: Span$\{u,v\}$ as a plane through the origin